

# Development of a Deep Reinforcement Learning Agent for Optimal Project Portfolio Budgeting under Uncertain Cash Flows

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## Abstract

The path toward a deployable decision-support system for project portfolio management comprises three steps: establishing domain foundations, formulating a tractable decision model, and validating at scale within incumbent systems. The first step is well served by existing literature — contract mechanics, earned value management, S-curve profiles, and portfolio selection models are mature — and the third remains future work. This paper occupies the second step through a relaxation hierarchy applied to a problem absent from the formal literature: *given a fixed amount of capital and a committed portfolio of contracted projects, how should a governing board allocate budget across projects at each discrete period to maximise discounted net cash flow while managing schedule deviation, FIDIC contract mechanics, and the option to terminate underperforming projects?*

**Level 1** formulates the problem as a deterministic full-foresight Mixed-Integer Linear Program whose optimum constitutes a provable upper bound no causal policy under uncertainty can exceed. **Level 2** introduces stochastic contractor performance, establishing that the resulting multi-stage stochastic program is intractable at realistic portfolio scales due to exponential scenario-tree growth. **Level 3** recasts the problem as a Constrained Markov Decision Process implemented as a configurable Gymnasium-compatible environment, on which a Proximal Policy Optimisation agent is trained and evaluated against the Level 1 oracle, a two-stage stochastic program, and a rule-based heuristic. The environment — encoding advance payment recovery, milestone payments, retention release, deviation penalties, and a two-condition termination gate — constitutes the primary contribution, positioned as the tractable middle step between established domain foundations and future large-scale deployment.

**Keywords:** project portfolio management; budget allocation under uncertainty; relaxation hierarchy; constrained Markov decision process; reinforcement learning; FIDIC contracts; earned value management; mixed-integer linear programming.

## 1 Introduction

### 1.1 Motivation and Problem Context

Large infrastructure and engineering projects routinely overrun their budgets and miss their schedules. Across a database of thousands of projects spanning seven decades, [Flyvbjerg et al. \(2003\)](#) document that cost overruns are the norm rather than the exception,

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with mean overruns exceeding 20% in transportation infrastructure and reaching 70% in large-scale IT systems. Merrow (2011) reports that fewer than 40% of industrial megaprojects achieve their approved cost and schedule targets. These figures describe individual projects. When a portfolio of such projects is executed simultaneously under a single governing body and a shared capital commitment, the risks compound: a cash-flow shortfall on one project propagates immediately to others, and the decision of how much budget to direct to each project at each period can determine whether the portfolio as a whole succeeds or fails.

Despite this, the formal treatment of the portfolio-level budget allocation problem as a sequential decision-making problem is conspicuously absent from the project portfolio management literature. Research attention has been concentrated at the *selection* stage — which projects to include in the portfolio — while the *operational governance* stage — how to manage the portfolio of committed projects through to completion — has received comparatively little analytical treatment (Archer and Ghasemzadeh, 1999; Cooper et al., 2001; Turner, 2014). The present work fills this gap.

We consider a portfolio of  $n$  projects, each governed by a bilateral contract consistent with FIDIC forms (Fédération Internationale des Ingénieurs-Conseils, 2017), executing concurrently under a fixed capital commitment. A portfolio manager — a board, investment committee, or programme director — exercises a single recurring decision at each discrete planning period: how much budget to allocate to each active project. All strategic and operational outcomes of interest flow through this allocation decision. The formal problem is to find the allocation policy that maximises discounted net cash flow across the portfolio horizon while satisfying hard liquidity constraints, respecting contract mechanics.

## 1.2 The Project Management Decision Hierarchy

To situate the present contribution precisely, it is useful to distinguish the three nested decision layers that constitute project portfolio management (Project Management Institute, 2017).

**Portfolio selection.** The selection problem asks: given a candidate set of projects and a capital budget, which subset should the organisation commit to? This is a combinatorial optimisation problem studied extensively in the literature using mean-variance portfolio theory (Markowitz, 1952; Eilat et al., 2006), multi-criteria decision analysis (Martinsuo and Lehtonen, 2007), and metaheuristic methods (Zhang et al., 2020). The present work treats selection as an exogenous input: the portfolio of committed projects is fixed at the start of the planning horizon, and contracts have already been signed. This decomposition is consistent with the two-stage governance model of Turner (2014) and the PMI Standard for Portfolio Management (Project Management Institute, 2017).

**Portfolio governance.** Once a portfolio is committed, the governing body must monitor and control execution. This is the layer at which the present work operates. The governing body does not manage individual projects directly — that is the contractor’s role — but it controls the budget directed to each project at each period. This single continuous allocation decision is the sole instrument of the portfolio manager. Termination is not a discrete agent action but an endogenous environment mechanic demonstrating project failure.

**Project execution.** At the project level, the contractor controls workforce, materials, equipment, and subcontracting. The portfolio manager observes the outcomes of these decisions — cumulative progress, earned value, payment certifications — but does not control them. The efficiency with which a budget allocation is converted into physical progress is determined at the project level and is uncertain from the portfolio manager’s perspective.

This hierarchy is central to the modelling architecture: the formulation developed in Section 4 lives entirely at the portfolio governance layer, with project execution dynamics appearing only as parameters (deterministic at Level 1, stochastic at Levels 2 and 3).

### 1.3 Contract Mechanics and Cash-Flow Structure

The financial interaction between the portfolio manager (as client) and each contractor is governed by the contract. This paper parameterises all contract mechanics consistently with FIDIC Red, Yellow, and Silver Book forms ([Fédération Internationale des Ingénieurs-Conseils, 2017](#)), which represent the dominant international framework for EPC contracting. Understanding the cash-flow structure is essential to understanding the optimisation problem; this subsection provides that foundation.

**Milestone-based payment.** Rather than paying contractors on a time-basis, FIDIC contracts structure payment around certified milestones: discrete progress thresholds at which the client verifies that a specified scope fraction has been physically completed and releases the corresponding payment. The gross payment at milestone  $j$  is a contractually fixed fraction  $\phi_{i,j}$  of the contract price  $CP_i$ . Because payment is contingent on certified progress, the timing of cash inflows is a function of the contractor’s actual performance and, through the portfolio manager’s allocation decisions, of the budget directed to the project. This creates the fundamental linkage between allocation decisions and cash inflows that makes the problem non-trivial.

**Advance payment and progressive recovery.** At project initiation, the client disburses an advance payment  $A_i = \alpha_i \cdot CP_i$  to the contractor. This is a timing instrument: it reduces the contractor’s early working-capital burden and is recovered progressively through deductions from subsequent milestone payments. The recovery is proportional to milestone payment weights, ensuring the full advance is recovered by final completion. The advance payment creates an immediate cash outflow at project start and a series of reduced inflows at each milestone, both of which must be tracked in the portfolio cash balance.

**Retention.** A fraction  $\rho_i$  of each milestone payment is withheld by the client as a performance bond. The accumulated retention  $R_i^{\text{ret}} = \rho_i \cdot CP_i$  is released in full upon project completion. Retention creates a structural completion incentive embedded in the payment mechanics: the larger the retention fraction, the more strongly the cash flow profile rewards completion over partial delivery. In the formulation, retention release enters the portfolio cash balance as a lump inflow in the period the final milestone is certified.

**Earned value management as a performance signal.** The Earned Value Management framework ([Fleming and Koppelman, 2010](#)) provides the diagnostic vocabulary used to monitor contractor performance. The Cost Performance Index  $CPI_{i,t} =$

$\text{BCWP}_{i,t}/\text{ACWP}_{i,t}$  measures how much budgeted value has been earned per unit of actual expenditure; values below unity indicate cost overrun. The Schedule Performance Index  $\text{SPI}_{i,t} = \text{BCWP}_{i,t}/\text{BCWS}_{i,t}$  measures actual progress relative to planned progress; values below unity indicate schedule lag (Fleming and Koppelman, 2010). In the present formulation, these indices are observable state variables that inform the allocation decision; their stochastic dynamics at Levels 2 and 3 are calibrated from empirical EVM datasets (Acebes et al., 2014; Batselier and Vanhoucke, 2017).

**S-curve expenditure profiles.** The planned expenditure of a construction or engineering project follows a characteristic sigmoid profile over its planned duration: slow ramp-up in the mobilisation phase, rapid expenditure in the peak execution phase, and tapering in the close-out phase (Kenley and Wilson, 2009). This profile is modelled using the regularized incomplete Beta function  $F_i(\tau) = I_\tau(a_i, b_i)$ , whose shape parameters  $(a_i, b_i)$  are calibrated from project data. The planned cumulative expenditure fraction at normalised time  $\tau_{i,t}$  provides the baseline against which actual progress is measured throughout the formulation.

**Termination.** FIDIC contracts provide for client-initiated termination when the contractor has materially breached the contract — most commonly through sustained failure to meet schedule obligations. Termination triggers a settlement payment based on the value of work completed at the termination date, net of any outstanding advance recovery obligation. In the present model, termination is a deliberate decision by the portfolio manager, not an exogenous event: it is triggered by the joint satisfaction of a cumulative penalty threshold and a sustained allocation floor, and its timing affects the portfolio cash balance through the settlement payment  $R_i^{\text{term}}$ , which may be positive or negative depending on the advance recovery position. Real options theory (Dixit and Pindyck, 1994; Huchzermeier and Loch, 2001) provides the theoretical basis for treating termination as the exercise of an abandonment option.

## 1.4 The Three-Level Modelling Arc

The formulation of Section 4 proceeds across three levels of increasing generality. This subsection provides the conceptual roadmap; each level is developed formally in its corresponding subsection.

**Level 1: Deterministic full-foresight MILP.** The natural starting point for any optimisation problem is its deterministic version: assume all uncertain parameters are known, and find the optimal decision sequence. In the portfolio budget allocation problem, the key uncertain parameter is the budget-to-progress efficiency  $\eta_{i,t}$  — the scalar that determines how much physical progress a given budget allocation yields in each period. If  $\eta_{i,t}$  is treated as a known constant  $\bar{\eta}_{i,t}$ , the progress recursion becomes a deterministic linear map from budget allocations to physical outcomes, and the entire optimisation problem collapses to a Mixed-Integer Linear Program (MILP).

The MILP is not proposed as a practical solution method — no real portfolio manager has advance knowledge of contractor efficiency trajectories. Rather, it serves two purposes. First, its optimal value  $Z_{L1}^*$  is a provable upper bound on the performance of any causal policy operating under uncertainty: since the MILP optimises with perfect foresight, it can do no worse than any policy that must make decisions without knowing future outcomes.

Second, it provides an LP foundation from which the stochastic extensions of Levels 2 and 3 are constructed by systematic relaxation of the deterministic assumption.

The MILP requires two structural choices that are standard in combinatorial optimisation. Milestone certification is a binary event — a milestone is either certified or not — captured by introducing binary decision variables  $u_{i,j,t} \in \{0,1\}$ . The symmetric running deviation penalty involves an absolute value and a cap, both of which are linearised by introducing auxiliary variables  $d_{i,t}^+, d_{i,t}^- \geq 0$  for the positive and negative parts of the deviation. These linearisations are exact: no approximation is introduced.

**Level 2: Stochastic extension and intractability.** When  $\eta_{i,t}$  is replaced by a random variable — as it must be to reflect reality — the MILP becomes a multi-stage stochastic program (Birge and Louveaux, 2011). The scenario-tree representation of this program grows exponentially: if  $\eta_{i,t}$  takes  $K$  discrete values at each of  $H$  periods for each of  $n$  projects, the scenario tree has  $K^{nH}$  leaf nodes. For a portfolio of ten projects over a 24-month horizon with five efficiency scenarios per project per period, this is  $5^{240}$  leaf nodes — a number that renders exact stochastic programming computationally intractable. This is not merely a computational inconvenience; it is the fundamental barrier that motivates the Level 3 reformulation.

**Level 3: Constrained MDP and RL agent.** The Constrained Markov Decision Process (Altman, 1999) provides the tractable reformulation. Rather than optimising over all scenario-tree paths simultaneously, the CMDP formalises the problem as a sequential interaction between a policy and an environment: at each period, the policy observes the current portfolio state, selects a budget allocation, and the environment transitions stochastically to the next state. The key insight is that the CMDP does not require explicit scenario enumeration: the policy is optimised through repeated interaction with the environment, using sampled trajectories rather than exhaustive scenario trees.

The mapping from MILP to CMDP is exact in structure. The decision variables  $\{x_{i,t}\}$  become the action space; the state variables ( $P_{i,t}^{\text{actual}}, E_{i,t}, \text{CPI}_{i,t}, \text{SPI}_{i,t}, \delta_{i,t}, \Delta_{i,t}^{\text{pen}}, \text{status}_{i,t}, B_t$ ) become the observation space; the budget constraint becomes the constrained action space; and the objective function becomes the reward signal. The CMDP is implemented as a configurable Gymnasium-compatible environment (Towers et al., 2024), and a Proximal Policy Optimisation agent (Schulman et al., 2017) is trained on it using domain randomisation (Tobin et al., 2017) to ensure generalisation across project types and contract structures.

## 1.5 The Operations Research Lineage

The present work is situated squarely within the Industrial Engineering and Operations Research tradition. The mathematical tools employed — mixed-integer linear programming, stochastic programming, dynamic programming, and Markov decision processes — share a common intellectual lineage that traces directly to Dantzig’s simplex method (Dantzig, 1963), Bellman’s principle of optimality (Bellman, 1957), and the stochastic programming framework of Dantzig (1955).

The MDP formalisation used in Level 3 is, mathematically, a computationally tractable approach to solving a Bellman equation in a high-dimensional state space. The state space in the present problem —  $7n + 1$  continuous and categorical variables for a portfolio of  $n$  projects — is too large for tabular dynamic programming and too structured for model-free value-iteration methods, but well-suited to policy-gradient methods that learn a parameterised policy directly. PPO (Schulman et al., 2017) is the policy-gradient algorithm of

choice for its stability properties in continuous action spaces and its established use in operations management domains ([Giannoccaro and Pontrandolfo, 2002](#); [Peng and Huang, 2021](#)).

The reinforcement learning agent is therefore not presented as a departure from the OR tradition, but as its continuation into the regime where the curse of dimensionality makes analytical dynamic programming intractable. The Level 1 MILP and Level 2 stochastic program are not straw men; they are the canonical OR solutions to the deterministic and stochastic versions of the problem, respectively, and both serve as baselines against which the Level 3 agent is evaluated.

## 1.6 Practical Deployment Context

Incumbent project management information systems — most prominently Oracle Primavera P6 and Microsoft Project — provide rich project-level schedule and cost tracking but lack a portfolio-level optimisation layer. The portfolio manager’s recurring budget allocation decision is currently supported, at best, by spreadsheet models, earned value dashboards, and expert judgement. The environment developed in Level 3 constitutes the analytical engine that such a portfolio-level decision-support module would require: it integrates contract mechanics, performance uncertainty, and multi-project cash-flow dynamics in a single configurable framework. This positions the contribution not only as a theoretical advance but as a step toward a deployable module for portfolio management information systems, a direction explicitly identified as a priority in the project management software literature ([Project Management Institute, 2017](#)).

## 1.7 Paper Organisation

The remainder of the paper is organised as follows. Section ?? reviews the four literature streams on which the work draws: project management foundations, project portfolio management and selection, PPM budgeting and cash-flow modelling, and reinforcement learning for sequential resource allocation. Section 4 develops the three-level problem formulation: the deterministic MILP (Level 1), the stochastic extension and intractability argument (Level 2), and the CMDP reformulation and Gymnasium-compatible environment (Level 3). Section ?? presents the empirical evaluation: calibration of environment parameters, training protocol, and comparison of the PPO agent against the Level 1 MILP oracle, a two-stage stochastic program, and a rule-based heuristic. Section ?? concludes with implications for portfolio management practice, limitations of the framework, and directions for future research including integration with incumbent PMIS tools.

# 2 Literature Review

## 2.1 Overview and Review Strategy

This section maps the intellectual lineage of the present work across four adjacent knowledge domains: project management (PM) foundations, project portfolio management (PPM) and selection, PPM budgeting and cash-flow modelling, and reinforcement learning (RL) applied to sequential resource-allocation problems. Together these streams define both the theoretical foundation and the gap that motivates the proposed framework. The review follows the PRISMA 2020 reporting guidelines ([Page et al., 2021](#)) to ensure transparency and reproducibility in source identification, screening, and inclusion decisions.

### 2.1.1 Search and Screening Protocol

Literature was retrieved from Google Scholar, Semantic Scholar, and Scopus using a progressively narrowing search strategy. Queries began at the broadest relevant domain (*project management*) and were successively refined through *project portfolio management*, *portfolio budgeting and cash-flow optimisation*, *earned value management under uncertainty*, and *reinforcement learning for project scheduling and resource allocation*. Citation-network traversal — backward and forward snowballing from high-citation anchor papers — was used to recover seminal works not surfaced by keyword search.

Records were screened in three sequential stages consistent with the PRISMA 2020 flow diagram (Appendix A). **Stage 1** (identification) assessed machine readability and bibliographic completeness. **Stage 2** (relevance filtering) applied cosine-similarity ranking against the research hypothesis using sentence-transformer embeddings, retaining sources above a calibrated relevance threshold. **Stage 3** (structured extraction) classified each retained paper by contribution type, key variables, modelling paradigm, and empirical basis. The PRISMA flow diagram quantifying records at each stage is provided in Appendix A.

## 2.2 Project Management: Foundational Concepts

The academic study of project management draws on schedule control (CPM/PERT), earned value management (EVM), and risk modelling as its core quantitative pillars (Fleming and Koppelman, 2010; Project Management Institute, 2021). EVM introduced the Cost Performance Index (CPI) and Schedule Performance Index (SPI) as real-time diagnostic signals relating budgeted cost of work performed to actual expenditure and planned schedule — providing the performance vocabulary that the present model inherits directly. Values of  $CPI < 1$  signal cost overrun;  $SPI < 1$  signals schedule lag. Stochastic extensions of EVM (Acebes et al., 2014; Batselier and Vanhoucke, 2017) and Monte Carlo simulation frameworks (Kwak and Ingall, 2007) have modelled performance uncertainty at the project level, establishing that CPI and SPI follow empirically characterisable distributions over project life.

S-curve analysis, formalised by Kenley and Wilson (2009) for construction cash flows, established that budget consumption follows characteristic sigmoid profiles whose shape parameters are project-type- and contract-form-dependent. The regularized incomplete Beta function provides a flexible single-family parameterisation of this profile (Barraza et al., 2004), and is adopted directly in the planned expenditure model of the present formulation.

Large-scale empirical evidence on project performance is provided by Merrow (2011) through the Independent Project Analysis database, and by Flyvbjerg et al. (2003); Flyvbjerg (2017) through infrastructure cost-overrun datasets spanning decades and geographies. These datasets inform both the calibration of uncertainty distributions in the present model and the benchmarks against which synthetic environment parameters are validated. Complementary empirical benchmarks are drawn from AACE International and Construction Industry Institute cost-engineering databases (Christensen, 1993).

Contract governance structures — particularly FIDIC forms (Red, Yellow, and Silver Books) — prescribe the financial interaction between owner and contractor through milestone payment schedules, advance payment and recovery mechanisms, and retention clauses (Fédération Internationale des Ingénieurs-Conseils, 2017). These mechanisms translate directly into the cash inflow and outflow dynamics formulated in Section 4.

## 2.3 Project Portfolio Management: Selection and Governance

Project portfolio management formalises the problem of allocating an organisation’s finite resources across a competing set of candidate projects to maximise strategic value (Archer and Ghasemzadeh, 1999; Cooper et al., 2001). The literature distinguishes two nested decision layers: *portfolio selection* — the binary or mixed-integer optimisation of which projects to include — and *portfolio governance* — the ongoing operational oversight of approved, active projects.

Portfolio selection models have been extensively developed. Markowitz-style mean–variance frameworks adapted from financial portfolio theory (Markowitz, 1952) have been applied to project baskets to trade off expected NPV against variance (Eilat et al., 2006). Multi-criteria decision analysis extensions incorporate strategic alignment, risk ratings, and resource interdependencies (Martinsuo and Lehtonen, 2007; Killen et al., 2008). Meta-heuristic approaches — genetic algorithms, particle swarm optimisation — address large combinatorial instances (Zhang et al., 2020). Resource-constrained portfolio selection under capital rationing has been studied by Kavadias and Loch (2003), whose no-borrowing constraint is adopted directly in the present formulation.

The present work *explicitly adopts the output of portfolio selection as an exogenous input*. The selection problem is treated as solved upstream; the focus is entirely on the operational governance layer that receives a committed portfolio and must manage it through execution. This decomposition is consistent with the two-stage governance model of Turner (2014) and the PMI Standard for Portfolio Management (Project Management Institute, 2017), which separates strategic selection from operational execution oversight. In doing so, this work builds directly on the selection literature while advancing the frontier into the governance domain that has received substantially less formal analytical treatment.

## 2.4 PPM Budgeting, Cash-Flow Modelling, and Uncertainty

Within portfolio governance, budget allocation across active projects over time represents a high-frequency, high-stakes operational decision. Despite its practical centrality, this specific problem has received limited formal treatment relative to portfolio selection. Existing work addresses project-level cash-flow modelling (Kenley and Wilson, 2009; Park and Peña-Mora, 2004), EVM-based forecasting of individual project cost-at-completion (Kim and Ballard, 2002; Acebes et al., 2014), and multi-project resource levelling (Lova et al., 2000; Van Peteghem and Vanhoucke, 2010). However, these contributions treat budgets as deterministic constraints rather than as dynamic decision variables subject to portfolio-level sequential optimisation.

Cash-flow uncertainty in construction and engineering projects arises from contractor performance variability, payment timing mismatches, scope changes, and external market shocks (Odeh and Battaineh, 2002; Doloi, 2012). Stochastic programming (Birge and Louveaux, 2011) provides the canonical OR framework for optimisation under such uncertainty, and its two-stage variant has been applied to project scheduling under resource uncertainty (Lova et al., 2000). However, multi-stage stochastic programs for portfolio-level sequential budget allocation face scenario-tree explosion as the number of projects and periods increases, as established formally in Section ??.

The gap in the literature is precise: *no established framework models the portfolio-level budget allocation decision as a sequential stochastic optimisation problem in which a central decision-maker allocates funds across multiple active projects whose cash inflow and outflow trajectories are uncertain and coupled through a shared capital constraint*. The present work addresses this gap.

## 2.5 Performance Recovery Interventions and Project Termination

Two operational levers available to a portfolio-level decision-maker beyond budget allocation are management interventions to recover underperforming projects and the exercise of the contractual termination option. Both have received treatment in the project management literature, but neither has been integrated into a portfolio-level sequential optimisation model.

Recovery interventions — crashing, scope reduction, procurement restructuring — are discussed in EPC and construction project management contexts (Meredith and Mantel, 2011; Pinto, 2016). The decision of when to intervene depends on real-time CPI and SPI signals and on the expected marginal value of recovery effort relative to its cost — a fundamentally sequential decision under uncertainty that the CMDP formulation in Level 3 is equipped to address.

Project termination as a formal decision problem has been studied through the lens of escalation of commitment (Staw and Ross, 1987) — the well-documented behavioural tendency to continue investing in failing projects beyond the economically rational stopping point — and through real options theory (Dixit and Pindyck, 1994; Huchzermeier and Loch, 2001). Real options models frame termination as the exercise of an abandonment option when continuation NPV falls below salvage value, a framing directly compatible with the two-condition termination gate of the present formulation. The penalty-saturation condition of the termination gate corresponds to the passage of the NPV threshold; the sustained allocation-floor condition confirms that the exercise decision is deliberate rather than transient.

## 2.6 Reinforcement Learning for Sequential Resource Allocation

Reinforcement learning provides a principled framework for sequential decision-making under uncertainty (Sutton and Barto, 2018). The MDP formalisation — state, action, transition, reward — maps naturally onto the portfolio budget allocation problem, and the constrained MDP extension (Altman, 1999) accommodates the hard budget feasibility constraints that the portfolio manager must satisfy at every period.

RL has been applied to project scheduling (Peng and Huang, 2021; Liu et al., 2023), resource-constrained project scheduling problems (RCPSP) (Tassel et al., 2023), and supply chain inventory management (Giannoccaro and Pontrandolfo, 2002). In each domain, RL demonstrates advantages over classical OR methods when state spaces are large, transition dynamics are stochastic, and the problem is non-stationary over the decision horizon. These are precisely the characteristics of the portfolio budget allocation problem at Levels 2 and 3.

Proximal Policy Optimisation (Schulman et al., 2017) is adopted as the solution algorithm for its stability in continuous action spaces, its sample efficiency relative to earlier policy-gradient methods, and its established use in operations management domains. Domain randomisation (Tobin et al., 2017) — randomising environment parameters during training to improve out-of-sample policy robustness — is adopted to ensure the learned policy generalises across the range of project types, contract structures, and market conditions represented in the calibration data. Walk-forward validation (Cerqueira et al., 2020) is used as the empirical evaluation protocol to guard against data leakage given the time-series structure of project cash-flow data.

Direct application of RL to portfolio-level budget allocation remains absent from the literature. The present work provides the first formulation of this problem as a CMDP and the first empirical demonstration that a PPO agent can learn non-trivial allocation

policies in a calibrated synthetic environment.

## 2.7 Synthesis: Positioning the Present Contribution

Table 1 maps representative works across the four literature streams against the key modelling dimensions of the present framework. Three structural gaps emerge.

First, the transition from portfolio *selection* to portfolio *operational governance* has not been modelled as a sequential decision problem. Selection literature assumes away post-selection management; project scheduling literature ignores portfolio interdependencies and the shared capital constraint.

Second, existing stochastic models of project cash-flow uncertainty operate at the individual project level and do not feed uncertainty upward into a portfolio-level decision architecture with a shared budget constraint.

Third, the endogenous mechanics of project failure — the joint modelling of cumulative schedule penalty, contractual floor allocation, and termination settlement — have not been embedded within a portfolio-level sequential decision framework in which budget allocation is the sole control variable.

These gaps are not isolated oversights; they reflect an incomplete progression along the path toward a deployable portfolio-level decision-support system. Figure 1 situates this progression as three sequential steps. **Step 1** — domain grounding — is well established by the literature reviewed above: contract mechanics, EVM performance signals, S-curve expenditure profiles, and portfolio selection models collectively provide the inputs and boundary conditions that any operational governance model requires. **Step 2** — tractable problem formulation — is the gap. Bridging it requires a relaxation hierarchy: the governance problem must first be stated precisely under idealised assumptions (a deterministic MILP establishing a performance ceiling), then shown to be intractable under realistic stochastic assumptions (exponential scenario-tree growth), and finally reformulated into a tractable approximation (a CMDP solved by a policy-gradient agent). No existing work completes this hierarchy at the portfolio level. **Step 3** — validated deployment within incumbent project management information systems such as Oracle Primavera P6 — remains future work, contingent on the tractable formulation that Step 2 must first provide.

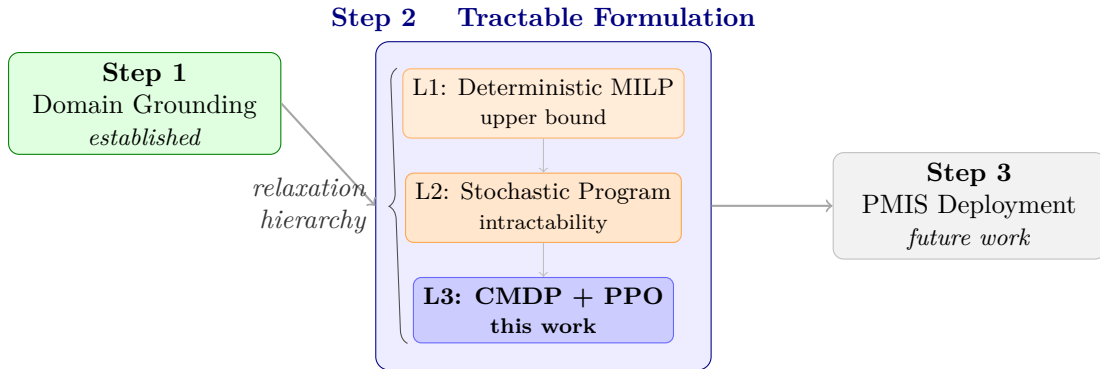


Figure 1: Solution maturity progression toward deployable portfolio-level budget allocation.

The present work addresses all three gaps and completes Step 2 of this progression through the relaxation hierarchy described above and developed formally in Section 4. The Gymnasium-compatible CMDP environment — with parameterisable contract mechanics,

stochastic performance model and termination — constitutes the primary methodological contribution. The PPO agent demonstrates its utility and provides the basis for empirical evaluation against classical OR baselines in Section ??.

Table 1: Representative literature mapped against modelling dimensions of the present framework. *Decision type*: S = selection, Sc = scheduling, B = budgeting. ✓ = fully addressed; ◦ = partially addressed; – = not addressed.

| Work                                  | Portfolio level | Decision type | Sequential decisions | Stochastic cash flows | Contract mechanics | Termination settlement |
|---------------------------------------|-----------------|---------------|----------------------|-----------------------|--------------------|------------------------|
| Eilat et al. (2006)                   | ✓               | S             | –                    | –                     | –                  | –                      |
| Klerides and Hadjiconstantinou (2019) | ✓               | S             | –                    | ◦                     | –                  | –                      |
| Huchzermeier and Loch (2001)          | ◦               | S             | ✓                    | ✓                     | –                  | ✓                      |
| Kenley and Wilson (2009)              | –               | B             | –                    | ◦                     | ◦                  | –                      |
| Acebes et al. (2014)                  | –               | B             | ◦                    | ✓                     | –                  | –                      |
| Lova et al. (2000)                    | ✓               | Sc            | –                    | –                     | –                  | –                      |
| Peng and Huang (2021)                 | –               | Sc            | ✓                    | ◦                     | –                  | –                      |
| Tassel et al. (2023)                  | –               | Sc            | ✓                    | –                     | –                  | –                      |
| <b>Present work</b>                   | ✓               | <b>B</b>      | ✓                    | ✓                     | ✓                  | ✓                      |

### 3 Problem Formulation

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### 4 Problem Formulation

This section develops the formal problem structure for portfolio-level budget allocation in contracted project portfolios. The formulation proceeds across three levels. **Level 1** (Section 4.2) treats all contractor performance parameters as known and establishes the deterministic MILP whose optimum constitutes a theoretical upper bound: no implementable policy operating under uncertainty can exceed it. **Level 2** (Section ??) reintroduces stochastic efficiency and characterizes the resulting multi-stage stochastic program, establishing that its scenario tree grows exponentially in the number of projects and periods. **Level 3** (Section ??) reformulates the intractable stochastic program as a Constrained Markov Decision Process (CMDP), providing an explicit mapping from LP elements to MDP primitives and yielding a Gymnasium-compatible environment on which the PPO agent of Section ?? is trained. The three levels are a coherent arc — Level 1 defines the ceiling, Level 2 explains why it is unreachable, Level 3 provides the tractable path — all sharing the primitives defined next.

#### 4.1 Scope, Assumptions, and Problem Primitives

**Scope.** The problem is operational, not strategic: the portfolio of contracted projects is fixed, commitments are signed, and the decision-maker exercises a single lever — the budget  $x_{i,t}$  directed to each active project  $i$  at each discrete period  $t$ . All outcomes of interest (milestone achievement, cash balance, schedule adherence, termination) flow through this allocation decision. Portfolio selection is treated as exogenous Archer and Ghasemzadeh (1999).

**Structural assumptions.** Four choices underlie the entire formulation. (i) *Independent bilateral contracts*: projects interact only through the shared budget constraint; no resource or workforce sharing occurs, consistent with standard EPC contracting practice. (ii) *Budget as the sole control*: the contractor’s internal resource decisions are absorbed into the efficiency parameter  $\eta_{i,t}$ ; the portfolio manager observes only how much physical progress a budget allocation yields, not how it is achieved. (iii) *Discrete monthly periods*:  $\mathcal{T} = \{1, \dots, H\}$  reflects the monthly governance cycle of portfolio management (progress reporting, payment certification, performance review). (iv) *FIDIC contract mechanics*: the milestone payment structure, advance payment and recovery, retention, and termination settlement are parameterized consistently with FIDIC Red, Yellow, and Silver Book forms [Fédération Internationale des Ingénieurs-Conseils \(2017\)](#); all contract-specific quantities are calibrated parameters, not fixed constants.

#### 4.1.1 Sets, Indices, and the Decision Variable

|                                  |                                                                                                                                                                         |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\mathcal{P}$                    | Projects $\{1, \dots, n\}$ , indexed by $i$ ; all committed at horizon start.                                                                                           |
| $\mathcal{T}$                    | Planning periods $\{1, \dots, H\}$ , indexed by $t$ (calendar months).                                                                                                  |
| $\mathcal{M}_i$                  | Ordered milestones $\{1, \dots, M_i\}$ for project $i$ , indexed by $j$ , with thresholds $\theta_{i,1} < \dots < \theta_{i,M_i} = 1$ .                                 |
| $\mathcal{A}_i^{\text{plan}}$    | Planned execution window $\{s_i, \dots, f_i\}$ ; anchors S-curve computations.                                                                                          |
| $\mathcal{P}_t^{\text{active}}$  | Projects active at period $t$ ; fixed parameter at Level 1, endogenous at Levels 2–3.                                                                                   |
| $\mathcal{M}_i^{\text{cert}}(t)$ | Milestones of project $i$ certified by period $t$ : $\{j : \theta_{i,j} \leq P_{i,t}^{\text{actual}}\}$ .                                                               |
| $x_{i,t} \geq 0$                 | <b>Decision variable</b> : budget allocated to project $i$ at period $t$ . The period- $t$ action is $\mathbf{x}_t = (x_{i,t})_{i \in \mathcal{P}_t^{\text{active}}}$ . |
| $B_t \geq 0$                     | Portfolio cash balance at the start of period $t$ ; $B_1$ is given.                                                                                                     |

#### 4.1.2 Project Parameters

Tables 2 and 3 list all project parameters, fixed at commitment. The planned cumulative expenditure fraction follows a normalized S-curve [Kenley and Wilson \(2009\)](#); [Barraza et al. \(2004\)](#) parameterized by  $(a_i, b_i)$ :

$$P_{i,t}^{\text{plan}} = F_i(\tau_{i,t}^c) = I_{\tau_{i,t}^c}(a_i, b_i), \quad \tau_{i,t} = \frac{t - s_i + 1}{D_i^{\text{plan}}}, \quad \tau_{i,t}^c = \min\{1, \max\{0, \tau_{i,t}\}\},$$

where  $I_\tau(\cdot, \cdot)$  is the regularized incomplete Beta function. The planned period expenditure increment is  $\Delta \bar{E}_{i,t} = \text{BAC}_i [F_i(\tau_{i,t}^c) - F_i(\tau_{i,t-1}^c)]$ . This profile is a fixed parameter derived at commitment from  $(a_i, b_i, s_i, f_i, \text{BAC}_i)$ ; it is never a decision variable.

| Parameter                                     | Symbol              | Domain           | Definition                                                                                         |
|-----------------------------------------------|---------------------|------------------|----------------------------------------------------------------------------------------------------|
| <b>Planned start &amp; finish</b><br>Schedule | $s_i, f_i$          | $\mathbb{Z}^+$   | Anchor of execution window $\mathcal{A}_i^{\text{plan}}$ and S-curve.                              |
| <b>Planned duration</b><br>Schedule           | $D_i^{\text{plan}}$ | $\mathbb{Z}^+$   | $f_i - s_i + 1$ ; normalizes progress against plan.                                                |
| <b>S-curve shape</b><br>Schedule              | $a_i, b_i$          | $\mathbb{R}^+$   | Beta CDF parameters governing the planned expenditure profile.                                     |
| <b>Milestone threshold</b><br>Schedule        | $\theta_{i,j}$      | $(0, 1]$         | Cumulative progress fraction required to certify milestone $j$ .                                   |
| <b>Delay buffer</b><br>Schedule               | $\text{buffer}_i$   | $\mathbb{Z}_0^+$ | Periods beyond $f_i$ before forced termination is assessed.                                        |
| <b>Contract price</b><br>Contract             | $CP_i$              | $\mathbb{R}^+$   | Total client payment; $CP_i = \text{BAC}_i(1 + \pi_i)$ .                                           |
| <b>Profit margin</b><br>Contract              | $\pi_i$             | $\mathbb{R}_0^+$ | Contractual markup on $\text{BAC}_i$ .                                                             |
| <b>Advance payment ratio</b><br>Contract      | $\alpha_i$          | $[0, 1]$         | $A_i = \alpha_i \cdot CP_i$ disbursed at $t = s_i$ ; recovered via milestone deductions.           |
| <b>Retention ratio</b><br>Contract            | $\rho_i$            | $[0, 1]$         | Fraction withheld per milestone; released as $R_i^{\text{ret}} = \rho_i \cdot CP_i$ at completion. |
| <b>Milestone weight</b><br>Contract           | $\phi_{i,j}$        | $\mathbb{R}^+$   | Fraction of $CP_i$ at milestone $j$ ; $\sum_j \phi_{i,j} = 1$ .                                    |

Table 2: Schedule and contract parameters for project  $i \in \mathcal{P}$  (fixed at commitment).

| Parameter                                   | Symbol      | Domain           | Definition                                                                                                                                                                       |
|---------------------------------------------|-------------|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Budget at completion</b><br>Cost         | $BAC_i$     | $\mathbb{R}^+$   | Total planned contractor expenditure; EVM cost basis.                                                                                                                            |
| <b>Discount factor</b><br>Cost              | $\gamma$    | $(0, 1]$         | Per-period discount applied to future cash flows; shared across $\mathcal{P}$ .                                                                                                  |
| <b>Initial cash balance</b><br>Cost         | $B_1$       | $\mathbb{R}^+$   | Portfolio cash balance at horizon start; initial condition of balance recursion.                                                                                                 |
| <b>Deviation penalty weight</b><br>Penalty  | $\lambda_i$ | $\mathbb{R}_0^+$ | Per-period cost per unit absolute progress deviation; calibrated so cumulative penalty at planned finish under sustained contractual-limit delay equals $\Lambda_i \cdot CP_i$ . |
| <b>Deviation penalty cap</b><br>Penalty     | $\Lambda_i$ | $\mathbb{R}_0^+$ | Maximum cumulative penalty as fraction of $CP_i$ ; applies symmetrically to lag and acceleration.                                                                                |
| <b>Termination threshold</b><br>Termination | $\kappa_i$  | $(0, 1]$         | Fraction of $\Lambda_i \cdot CP_i$ at which project enters termination candidacy.                                                                                                |
| <b>Minimum allocation</b><br>Termination    | $m_i^0$     | $\mathbb{R}_0^+$ | Floor budget to hold project in contractually serviceable state while termination conditions are evaluated; no performance-recovery role.                                        |
| <b>Confirmation window</b><br>Termination   | $k$         | $\mathbb{Z}^+$   | Consecutive periods at floor $m_i^0$ required to confirm abandonment.                                                                                                            |

Table 3: Cost, penalty, and termination parameters for project  $i \in \mathcal{P}$  (fixed at commitment).

#### 4.1.3 State Variables and Transition Dynamics

**Project-level state.** For each  $i \in \mathcal{P}$ ,  $t \in \mathcal{T}$ :  $P_{i,t}^{\text{actual}} \in [0, 1]$  (cumulative progress),  $E_{i,t} \geq 0$  (cumulative expenditure),  $SPI_{i,t} = P_{i,t}^{\text{actual}}/P_{i,t}^{\text{plan}}$ ,  $CPI_{i,t} = P_{i,t}^{\text{actual}} \cdot BAC_i/E_{i,t}$ ,  $\delta_{i,t} = P_{i,t}^{\text{actual}} - P_{i,t}^{\text{plan}}$  (signed deviation),  $\Delta_{i,t}^{\text{pen}} \geq 0$  (cumulative penalty), and  $\text{status}_{i,t} \in \{\text{active}, \text{completed}, \text{terminated}\}$ . The portfolio state at period  $t$  is:

$$S_t = \left( \{P_{i,t}^{\text{actual}}, E_{i,t}, SPI_{i,t}, CPI_{i,t}, \delta_{i,t}, \Delta_{i,t}^{\text{pen}}, \text{status}_{i,t}\}_{i \in \mathcal{P}}, B_t \right).$$

**Budget-to-progress conversion.** Budget  $x_{i,t}$  converts to progress through the efficiency  $\eta_{i,t}$ :

$$P_{i,t}^{\text{actual}} = P_{i,t-1}^{\text{actual}} + \eta_{i,t} \cdot \frac{x_{i,t}}{BAC_i}, \quad E_{i,t} = E_{i,t-1} + x_{i,t}, \quad (1)$$

with  $P_{i,0}^{\text{actual}} = E_{i,0} = 0$ . The scalar  $\eta_{i,t}$  absorbs all sources of conversion uncertainty (productivity, supply chain, weather). At Level 1,  $\eta_{i,t}$  is replaced by the known deterministic value  $\bar{\eta}_{i,t}$  (equal to unity under nominal plan); at Levels 2–3 it is a stochastic primitive realized only after  $x_{i,t}$  is committed.

**Contractual cash-flow mechanics.** Payment inflows follow FIDIC mechanics [Fédération Internationale des Ingénieurs-Conseils \(2017\)](#). At initiation ( $t = s_i$ ), the advance

payment  $A_i = \alpha_i \cdot CP_i$  is received. Milestone  $j$  is certified at the first  $t$  with  $P_{i,t}^{\text{actual}} \geq \theta_{i,j}$ ; the net payment (after retention and advance recovery) is:

$$R_{i,j}^{\text{net}} = \phi_{i,j} \cdot CP_i(1 - \rho_i) - \phi_{i,j} \cdot A_i. \quad (2)$$

Upon completion ( $P_{i,t}^{\text{actual}} = 1$ ) the accumulated retention  $R_i^{\text{ret}} = \rho_i \cdot CP_i$  is released together with the final milestone payment. The payment identity  $A_i + \sum_j R_{i,j}^{\text{net}} + R_i^{\text{ret}} = CP_i$  holds exactly. Aggregate period inflow is:

$$V_t = \sum_{i,j} R_{i,j}^{\text{net}} \cdot \mathbf{1}\{(i,j) \text{ newly certified at } t\} + \sum_{i \in \mathcal{P}_t^{\text{term}}} R_i^{\text{term}}, \quad (3)$$

where  $\mathcal{P}_t^{\text{term}}$  collects projects newly terminated at  $t$ , with settlement  $R_i^{\text{term}}$  defined in (7) below.

**Symmetric running deviation penalty.** Schedule adherence is priced at every period:

$$c_{i,t}^{\text{dev}} = \min\{\lambda_i |\delta_{i,t}|, \Lambda_i \cdot CP_i\}, \quad \Delta_{i,t}^{\text{pen}} = \sum_{\tau=s_i}^t c_{i,\tau}^{\text{dev}}. \quad (4)$$

The penalty is symmetric ( $|\delta_{i,t}|$ , not  $\max\{0, -\delta_{i,t}\}$ ): both lag and acceleration are penalized, discouraging budget-pushing that risks quality degradation [Merrow \(2011\)](#). The cap  $\Lambda_i \cdot CP_i$  mirrors the contractual delay-penalty ceiling under FIDIC terms.

**Termination conditions and settlement.** A project transitions to **terminated** at the first period  $t$  at which both conditions hold simultaneously:

$$(i) \text{ Penalty saturation: } \Delta_{i,t}^{\text{pen}} \geq \kappa_i \cdot \Lambda_i \cdot CP_i, \quad (5)$$

$$(ii) \text{ Confirmed floor: } x_{i,\tilde{t}} \leq m_i^0 \quad \forall \tilde{t} \in \{t - k + 1, \dots, t\}. \quad (6)$$

Condition (5) signals exhaustion of contractual schedule tolerance; condition (6) confirms that the portfolio manager has deliberately ceased investment rather than transiently reduced it. Upon termination at  $t_i^{\text{term}}$ , the net settlement is:

$$R_i^{\text{term}} = P_{i,t_i^{\text{term}}}^{\text{actual}} \cdot CP_i(1 - \rho_i) - \left( A_i - \sum_{j \in \mathcal{M}_i^{\text{cert}}(t_i^{\text{term}})} \phi_{i,j} \cdot A_i \right). \quad (7)$$

Negative  $R_i^{\text{term}}$  (outstanding advance exceeds earned value) constitutes a net obligation recovered from portfolio inflows, consistent with FIDIC termination clauses.

**Portfolio cash balance recursion.**

$$B_{t+1} = B_t - \sum_{i \in \mathcal{P}_t^{\text{active}}} x_{i,t} + V_t, \quad B_t \geq 0 \quad \forall t \in \mathcal{T}. \quad (8)$$

The constraint  $B_t \geq 0$  prohibits borrowing; the portfolio operates under a fixed capital commitment [Kavadias and Loch \(2003\)](#). The shared balance  $B_t$  is the sole coupling across otherwise independent project dynamics, and its linear scaling with  $n$  characterizes the complexity that motivates the intractability argument in Level 2.

#### 4.1.4 Shared Objective

The decision-maker maximizes discounted net cash flow over the horizon:

$$\max_{\{x_{i,t}\}} \mathbb{E} \left[ \sum_{t=1}^H \gamma^{t-1} \left( V_t - \sum_{i \in \mathcal{P}_t^{\text{active}}} x_{i,t} - \sum_{i \in \mathcal{P}_t^{\text{active}}} c_{i,t}^{\text{dev}} \right) \right], \quad (9)$$

subject to, for all  $i \in \mathcal{P}$ ,  $t \in \mathcal{T}$ :

$$x_{i,t} \geq 0, \quad (10)$$

$$\sum_{i \in \mathcal{P}_t^{\text{active}}} x_{i,t} \leq B_t, \quad (11)$$

$$x_{i,t} \geq m_i^0 \cdot \mathbf{1}\{\text{status}_{i,t} = \text{active}\}. \quad (12)$$

There is no terminal sum. Completion incentives are embedded structurally:  $R_i^{\text{ret}}$  and  $R_{i,M_i}^{\text{net}}$  enter  $V_t$  at the certification period;  $R_i^{\text{term}}$  (possibly negative) enters at termination, creating a natural deterrent against premature abandonment. Constraint (12) ensures each active project remains in a contractually serviceable state; it is vacuous once  $\text{status}_{i,t} \neq \text{active}$ . The expectation  $\mathbb{E}[\cdot]$  collapses to a deterministic evaluation at Level 1 and is retained at Levels 2 and 3.

#### 4.2 Level 1: Deterministic Full-Foresight MILP

Setting  $\eta_{i,t} = \bar{\eta}_{i,t}$  (known) renders  $S_t$  a deterministic function of  $\mathbf{x}_t$ , so  $\mathcal{P}_t^{\text{active}}$ , milestone certification times, and termination periods  $t_i^{\text{term}}$  are all computable ex ante. Introducing the certification binary  $u_{i,j,t} \in \{0,1\}$  ( $= 1$  iff milestone  $j$  of project  $i$  certified by period  $t$ ) and activity indicator  $\text{act}_{i,t} \in \{0,1\}$  (fixed parameter), the stochastic objective (9) collapses to the MILP:

$$\max \sum_{t=1}^H \gamma^{t-1} \left( V_t^{\text{det}} - \sum_{i \in \mathcal{P}} \text{act}_{i,t} x_{i,t} - \sum_{i \in \mathcal{P}} \text{act}_{i,t} c_{i,t}^{\text{dev}} \right) \quad (13a)$$

$$\text{s.t. } B_{t+1} = B_t - \sum_i \text{act}_{i,t} x_{i,t} + V_t^{\text{det}}, \quad B_t \geq 0, \quad \forall t \quad (13b)$$

$$\sum_i \text{act}_{i,t} x_{i,t} \leq B_t, \quad \forall t \quad (13c)$$

$$P_{i,t}^{\text{actual}} = P_{i,t-1}^{\text{actual}} + \bar{\eta}_{i,t} \frac{x_{i,t}}{\text{BAC}_i}, \quad P_{i,t}^{\text{actual}} \leq 1, \quad \forall i, t \quad (13d)$$

$$P_{i,t}^{\text{actual}} \geq \theta_{i,j} u_{i,j,t} - \mathbf{M}(1 - u_{i,j,t}), \quad \forall i, j, t \quad (13e)$$

$$P_{i,t}^{\text{actual}} \leq \theta_{i,j} + \mathbf{M} u_{i,j,t}, \quad u_{i,j,t} \geq u_{i,j,t-1}, \quad u_{i,j+1,t} \leq u_{i,j,t}, \quad \forall i, j, t \quad (13f)$$

$$\begin{aligned} V_t^{\text{det}} = & \sum_{i,j} R_{i,j}^{\text{net}} (u_{i,j,t} - u_{i,j,t-1}) + \sum_i R_i^{\text{ret}} (u_{i,M_i,t} - u_{i,M_i,t-1}) \\ & + \sum_i A_i \cdot \mathbf{1}\{t=s_i\} + \sum_{i \in \mathcal{P}^{\text{term}}} R_i^{\text{term}} \cdot \mathbf{1}\{t=t_i^{\text{term}}\}, \quad \forall t \end{aligned} \quad (13g)$$

$$\delta_{i,t} = P_{i,t}^{\text{actual}} - P_{i,t}^{\text{plan}} = d_{i,t}^+ - d_{i,t}^-, \quad d_{i,t}^+, d_{i,t}^- \geq 0, \quad \forall i, t \quad (13h)$$

$$c_{i,t}^{\text{dev}} \geq \lambda_i (d_{i,t}^+ + d_{i,t}^-), \quad c_{i,t}^{\text{dev}} \leq \Lambda_i \cdot CP_i, \quad c_{i,t}^{\text{dev}} \geq 0, \quad \forall i, t \quad (13i)$$

$$x_{i,t} \geq m_i^0 \cdot \text{act}_{i,t}, \quad \forall i, t \quad (13j)$$

$$x_{i,t} \geq 0, \quad u_{i,j,t} \in \{0,1\}, \quad u_{i,j,0} = P_{i,0}^{\text{actual}} = 0. \quad (13k)$$

The big- $M$  constant in (13e)–(13f) is  $\mathbf{M} = \max_{i,j} \theta_{i,j}$ ; tighter values may be derived project-specifically. Constraints (13e)–(13f) implement the exact if-and-only-if certification logic with monotone, ordered binary activation. Constraints (13h)–(13i) linearize the absolute-value and cap in (4): the maximizing objective drives  $c_{i,t}^{\text{dev}}$  to the tighter of the two bounds, recovering the exact symmetric penalty. The difference  $(u_{i,j,t} - u_{i,j,t-1})$  in (13g) equals one in the sole period a milestone is first certified, linearizing the trigger without additional binaries. Termination timing  $t_i^{\text{term}}$  and the settlement  $R_i^{\text{term}}$  (7) are pre-computed from  $\bar{\eta}_{i,t}$  before solving; the termination conditions (5)–(6) are parameter-verifiable at Level 1 and become binding optimization constraints at Levels 2–3.

**Upper bound.** Let  $Z_{\text{L1}}^*$  denote the Level 1 optimum. For any causal policy  $\pi$  feasible under uncertainty,  $\sup_{\pi} \mathbb{E}[Z^{\pi}] \leq Z_{\text{L1}}^*$ : every feasible causal allocation sequence is also feasible in the Level 1 program solved on the realized trajectory, so the full-foresight optimum dominates in expectation. The gap  $Z_{\text{L1}}^* - Z^{\pi}$  quantifies the cost of uncertainty and anchors the empirical comparison in Section ??.

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